

The Deep-Regime Crossover as First-Passage Registration closing the sub-threshold dynamics of the weak-acceleration floor (ledger D4)

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Abstract

The weak-acceleration floor $a_0 = cH_0/2\pi$ is derived in the body as the rate at which the drive decorrelates links: a static gradient is signal only if it accumulates at least one phase cycle per Hubble time. This note closes the dynamics *below* that floor. A sub-floor gradient $g_N < a_0$ accumulates $x = g_N/a_0 < 1$ cycles per Hubble time; its coincidence amplitude is $A = \sqrt{x}$, and registration is the first passage of the decorrelating (killed) meridian walk to that amplitude barrier. The resolved fraction is $f = 1 - \eta^a$, with η the in-(0,1) root of the rational quadratic $w\eta^2 - 2\eta + w = 0$ ($w = 1 - s$ the per-chronon survival), whose resolvable-window reading is $f = 1 - e^{-\sqrt{x}}$. Flux conservation (the body’s Lemma fluxnoise) leaves the mean flux unchanged, so the observer attributes the conserved flux to the resolved fraction and infers

$$g_{\text{obs}} = \frac{g_N}{1 - e^{-\sqrt{g_N/a_0}}}.$$

This is the radial acceleration relation: deep limit $g_{\text{obs}} \rightarrow \sqrt{g_N a_0}$ (flat curves, the baryonic Tully–Fisher relation $v^4 = GMa_0$), Newtonian limit $g_{\text{obs}} \rightarrow g_N$. The mechanism the body left open — “no quantitative model of this dynamics yet exists” — is supplied, and it is the registration dynamics the body required, not a modification of the mean force. Every exact object is rational or quadratic-extension algebraic; the continuum readings are labelled [approx] and verified in `validation/validate_deepregime.py`.

1 The floor and the sub-floor count

The body establishes (Proposition `afloor`) that the drive decorrelates every link at the Hubble rate, so the resolution floor $1/\sqrt{\Omega}$ reads, as an acceleration, the threshold

$$a_0 = \frac{cH_0}{2\pi},$$

the gradient that accumulates exactly one phase cycle per Hubble time. A static baryonic gradient g_N therefore accumulates

$$x := \frac{g_N}{a_0}$$

phase cycles per Hubble time. For $g_N \geq a_0$ the signal clears the floor ($x \geq 1$) and the Newtonian reading of the body holds. For $g_N < a_0$ the signal is sub-floor ($x < 1$): it is not absent, but it is registered only intermittently, and the question the body leaves open is the law of that registration.

2 Registration as first passage

A sub-floor signal is registered as a coincidence count. The count accumulated against the floor is $N = x$, and by the coincidence (Born) reading of the programme the registrable quantity is the *amplitude* $A = \sqrt{N} = \sqrt{x}$, not the count itself [?]. Registration is then a first-passage event:

the meridian phase walk, decorrelated (killed) by the drive at per-chronon rate s , must reach the amplitude barrier A before it is killed.

Lemma 1 (Registration fraction). *Let the meridian walk survive each chronon with probability $w = 1 - s$ and step ± 1 symmetrically. The probability of reaching an amplitude barrier a chronon-steps away before being killed is η^a , where $\eta \in (0, 1)$ is the root of the rational quadratic*

$$w\eta^2 - 2\eta + w = 0, \quad \eta = \frac{1 - \sqrt{1 - w^2}}{w}.$$

The unregistered fraction is η^a and the resolved fraction is $f = 1 - \eta^a$.

Proof. The reach probability q_k from distance k satisfies the killed recurrence $q_k = w \cdot \frac{1}{2}(q_{k-1} + q_{k+1})$, whose decaying solution is $q_k \propto \eta^k$ with $w \frac{1}{2}(\eta + \eta^{-1}) = 1$, i.e. $w\eta^2 - 2\eta + w = 0$; the root in $(0, 1)$ is $\eta = (1 - \sqrt{1 - w^2})/w$. The barrier value $q_0 = \eta^a$ gives the reach probability. The coefficients are rational, so η is algebraic in the framework's quadratic extension; no continuum object enters. $\square$

Remark 2 (The resolvable-window reading). Writing $w = 1 - s$ and $1 - w^2 = 2s - s^2$, the decay constant is $-\ln \eta = \operatorname{arccosh}(1/w) = \sqrt{2s}(1 + \frac{s}{6} + \dots)$, so $\eta^a = e^{-a\sqrt{2s}}(1 + O(s))$. The barrier height and killing rate combine into the single invariant $a\sqrt{2s} = \sqrt{x}$ (the amplitude of the count x). Hence, in the resolvable window $s \rightarrow 0$ at fixed x ,

$$f = 1 - \eta^a \xrightarrow{[\text{approx}]} 1 - e^{-\sqrt{x}}.$$

This is exact on the substrate (Lemma 1); the exponential is its continuum reading.

3 The crossover

Theorem 3 (The registration crossover). *The mean synchronisation flux through any sphere is conserved (the body's Lemma fluxnoise): floor noise redistributes the transfer in time but not in mean. A bounded observer resolves only the fraction f of that flux as a coherent gradient, and by conservation attributes the whole conserved flux to the resolved fraction. The inferred acceleration is therefore*

$$g_{\text{obs}} = \frac{g_N}{f} = \frac{g_N}{1 - e^{-\sqrt{g_N/a_0}}}.$$

Remark 4 (Consistency with the mean-force constraint). Theorem 3 does not modify the mean static force, which Lemma fluxnoise forbids. The enhancement is a property of the *reading*: the registered gradient carries only the resolved fraction f of the conserved flux, and the observer's inference g_N/f is the amplification this registration imposes. This is exactly the combination the body's Remark **deepmond** demanded — a change in the reading of motion that preserves the mean flux. The dynamics is the killed-walk first passage of Lemma 1; the rotated-chart picture of the body is its conjugate-chart description, the resolved and unresolved weights being $\cos^2 \alpha = f$ and $\sin^2 \alpha = \eta^a \rightarrow e^{-\sqrt{x}}$.

4 Limits

Deep regime ($x \rightarrow 0$): $1 - e^{-\sqrt{x}} \rightarrow \sqrt{x}$, so

$$g_{\text{obs}} \rightarrow \frac{g_N}{\sqrt{g_N/a_0}} = \sqrt{g_N a_0},$$

the deep-MOND law. With $g_N = GM/r^2$ this gives the asymptotically flat speed $v^2 = r g_{\text{obs}} = \sqrt{GMa_0}$, hence the baryonic Tully–Fisher relation

$$v^4 = GMa_0.$$

Newtonian regime ($x \rightarrow \infty$): $f \rightarrow 1$, $g_{\text{obs}} \rightarrow g_N$. The crossover is the radial acceleration relation, with deep log-slope $\frac{1}{2}$ and Newtonian log-slope 1.

5 Verification

The script `validation/validate_deepregime.py` checks, float-free where exact: (i) the registration constant η satisfies $w\eta^2 - 2\eta + w = 0$ as an exact rational identity, and $\eta^a \rightarrow e^{-\sqrt{x}}$ to relative error $O(s)$ across $s \in \{1/50, \dots, 1/12800\}$ at fixed $\sqrt{x} = 0.6$ (relative error $5 \times 10^{-3} \rightarrow 2 \times 10^{-5}$); (ii) the crossover log-slopes are 0.500 (deep) and 1.000 (Newtonian); (iii) the deep limit reproduces $v^4 = GMa_0$; and (iv) the crossover is the McGaugh–Lelli–Schombert 2016 fitting function with $g_{\dagger} = a_0$, identical to relative 2×10^{-16} over twelve decades. The exponential and the empirical comparison are continuum readings, labelled [approx]; the registration constant and the count $x = g_N/a_0$ are exact.

Status. Ledger D4 moves from **O** to **T|B**: the sub-threshold dynamics is the first-passage registration of Lemma 1 (theorem), given the coincidence-amplitude reading $A = \sqrt{N}$ (bridge, [?]). The broader galactic phenomenology built on this crossover — intrinsic scatter as dynamical temperature, the external-field effect, and clusters — is developed in [?].